

Synthesis of Symmetrical and Unsymmetrical Functionalized Arylphosphines from Chlorophosphines and Organozinc Reagents

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Abstract: A stepwise procedure allowing the formation of symmetrical arylphosphines is described. It relies on the use of preformed functionalized aromatic organozinc reagents to perform arylations of chlorophosphines. Some preliminary results concerning the synthesis of unsymmetrical diarylphenylphosphines through sequential coupling of organozinc species with dichlorophenylphosphine are also reported.

Key words: phosphorus, arylation, organometallic reagents, chlorophosphines, functionalized arylphosphines

Arylphosphines are widely used as ligands in homogeneous catalysis. The presence of functional groups on aryl moieties can obviously offer the advantage of modulating the catalytic properties of the coordinated metallic center. Thus, in order to obtain more and more selective procedures, many efforts are directed towards the synthesis of highly functionalized triarylphosphines. Previous works dealing with the synthesis of diarylphenyl- and triarylphosphines involve the use of aromatic Grignard reagents¹ or aryllithium compounds.² Some limitations arise from these procedures. Experimental conditions are generally drastic and various functional groups can be incompatible with reaction conditions. On the contrary, functionalized aromatic organozinc reagents are easily accessible in mild conditions. In addition, a large variety of organozinc derivatives are readily available and due to their broad chemical tolerance they constitute an interesting source of nucleophiles.³ Several years ago, Knochel et al. reported the use of organozinc obtained from alkyl and aryl iodides for the nucleophilic displacement of chlorodiphenylphosphine.⁴

In a previous paper, we described the facile synthesis of functionalized aryldiphenylphosphines using a nickel-catalyzed reductive cross-coupling between chlorodiphenylphosphine and aryl bromides.⁵ Several attempts to extend this procedure to the synthesis of symmetrical diarylphenylphosphines and triarylphosphines starting, respectively, from dichlorophenylphosphine and trichlorophosphine were inefficient. Consequently, we envisaged a stepwise approach to these compounds in which preformed functionalized aromatic organozinc reagents would be coupled with chlorophosphines.

In this paper we report the mono-, di- and triarylation of chlorophosphines into symmetrical arylphosphines using preformed functionalized aromatic organozinc reagents. An interesting feature of this procedure is its adequacy in the synthesis of unsymmetrical arylphosphines through sequential addition of different organozinc reagents to chlorophosphines. Thus, some preliminary results in this area are also mentioned.

Organozinc reagents were obtained from aryl bromides according to a procedure previously developed in the laboratory.⁶ In order to test their reactivity, organozinc compounds were exposed to chlorodiphenylphosphine and allowed to react for a few hours at room temperature.⁷ Results are presented in Table 1.

Table 1 Synthesis of Aryldiphenylphosphines

Entry	FG	Product	Isolated yield (%) ^a
1	4-OMe	1a	45 (91) ^b
2	3-CN	1b	48
3	3-CO ₂ Et	1c	70
4	3-CF ₃	1d	72

^a Experimental conditions: ArZnBr (10–12 mmol) in MeCN and Ph₂PCl (10 mmol) were allowed to react for 2 h at r.t.

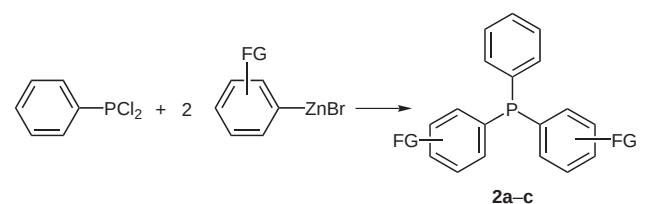
^b Isolated as its phosphine–borane complex. In order to realize this protection, CeCl₃·7H₂O (30 mmol) and NaBH₄ (30 mmol) were added to the mixture and allowed to react for 30 min at 0 °C.

Yields are generally good, ranging from 45–72%. It can be noted that limitations are essentially due to the oxidation of coupling products upon work up and chromatographic purification. Indeed, a protection of aryldiphenylphosphines as their phosphine–borane complexes⁸ using CeCl₃·7H₂O and NaBH₄⁹ on crude mixtures gave rise to the significant improvement of yields as attested by results reported in entry 1.

Having shown that the coupling step between organozinc compounds and chlorodiphenylphosphine was rather

efficient, we envisaged the diarylation of dichlorophenylphosphine in related conditions. Thus, two equivalents of functionalized organozinc compounds were allowed to react with dichlorophenylphosphine a few hours at room temperature. Results are reported in Table 2.

Table 2 Synthesis of Symmetrical Diarylphenylphosphines^a



Entry	FG	Product	Yield (%) ^b
1	4-OMe	2a	37
2	4-CO ₂ Et	2b	37
3	3-CF ₃	2c	38

^a Experimental conditions: ArZnBr (10–12 mmol) in MeCN and PhPCl₂ (5 mmol) were allowed to react for 2 h at r.t.

^b Isolated yields.

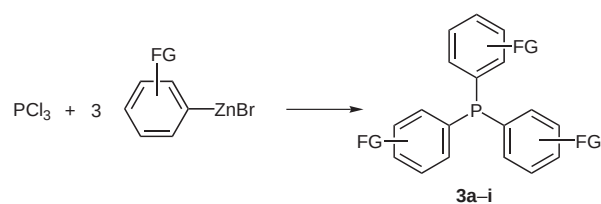
Results are moderate but homogeneous (37–38%). Chromatographic purifications proved to be difficult. Indeed, coupling products and biaryl compounds, arising from the dimerization of the starting aryl bromide during the organozinc synthesis step, exhibited very close *R_f* values. The protection of diarylphenylphosphines as their phosphine–borane complexes might improve the ease of chromatographic separation.

To supplement this work, we undertook to study the reactivity of organozinc compounds towards trichlorophosphine.

As shown in Table 3, results are good, either with electron-donating or electron-withdrawing groups connected to the phenyl moiety. The *para* or *meta* orientation of substituents seems to have a limited influence on the course of the reaction since similar yields are obtained in both cases.

In order to extend the procedure to the synthesis of unsymmetrical diarylphenylphosphines, two different organozinc compounds were prepared separately. It was assumed that the most reactive organozinc of both had to be added first. Indeed, the first arylation had to be as rapid as possible to ensure the complete conversion into arylchlorophenylphosphine before adding the second organozinc reagent. In addition, the first organozinc might be added in the exact stoichiometric amount¹⁰ to avoid the formation of symmetrical diarylated products. On the other hand, the second organozinc reagent was added in slight excess to favor the second substitution. Thus, considering the fact that in general, electron-rich organozinc are more reactive than electron-poor organozinc reagents, dichlorophenylphosphine was exposed sequentially to 4-

Table 3 Synthesis of Symmetrical Triarylphosphines^a



Entry	FG	Product	Yield (%) ^b
1	4-Me	3a	37
2	4-CO ₂ Et	3b	36
3	4-OMe	3c	45
4	4-CN	3d	66
5	4-CF ₃	3e	53
6	4-Cl	3f	34
7	3-Me	3g	65
8	3-OMe	3h	30
9	3-CF ₃	3i	65

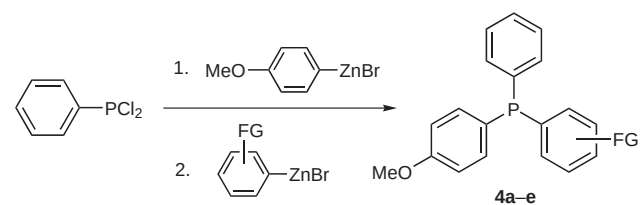
^a Experimental conditions: ArZnBr (10–12 mmol) in MeCN and PCl₃ (2.5 mmol) were allowed to react for 2 h at r.t.

^b Isolated yields.

methoxyphenylzinc bromide and some different organozinc compounds. Results are presented in Table 4.

Yields are moderate but encouraging since these results correspond to preliminary works. Purifications were found to be easier when diarylphenylphosphines had been

Table 4 Synthesis of Unsymmetrical Diarylphenylphosphines^a



Entry	FG	Product	Yield (%) ^b
1	4-CN	4a	15
2	4-CN	4b	26 ^c
3	2-Me	4c	36 ^c
4	3-CF ₃	4d	51 ^c
5	4-CO ₂ Et	4e	16 ^c

^a Experimental conditions: Ar¹ZnBr (10 mmol) in MeCN and PhPCl₂ (10 mmol) were allowed to react for 5 min at r.t. The second organozinc Ar²ZnBr (10–12 mmol) was then added and the mixture was stirred for additional 2 h at r.t.

^b Isolated yields.

^c Isolated as its oxide. In order to realize the oxidation, a slight excess of aqueous H₂O₂ was added to the mixture and allowed to react for 10 min at r.t.

totally oxidized using H_2O_2 before work up. It should be mentioned that symmetrical diarylphenylphosphines were detected only as traces in the reaction mixture. In order to improve products recovery, a protection of diarylphenylphosphines as phosphine–borane complexes would also be appropriate and perhaps even more efficient since deprotection steps are clearly easier than reduction steps. Nevertheless, we found rather difficult to realize the protection on crude mixtures without losing a part of the product under the oxide form.

In conclusion, the results reported in this paper show that aromatic organozinc reagents constitute valuable nucleophiles in the synthesis of functionalized symmetrical arylphosphines¹¹ starting from chlorophosphines. Couplings are achieved in very mild conditions compared to those commonly used with organolithium or Grignard reagents and selectivities are satisfactory. Unsymmetrical diarylphenylphosphines¹¹ can also be obtained using a sequential coupling between two different organozinc reagents and dichlorophenylphosphine. These preliminary results are encouraging and prompt us to optimize reactions and work-up conditions. In addition, sustained efforts are in progress to realize the synthesis of unsymmetrical triarylphosphines starting from trichlorophosphine.

References and Notes

- (1) (a) Mann, F. G.; Chaplin, E. J. *J. Chem. Soc.* **1937**, 527. (b) Hart, F. A.; Mann, F. G. *J. Chem. Soc.* **1955**, 4107. (c) Russell, M. G.; Warren, S. *Tetrahedron Lett.* **1998**, 39, 7995. (d) Whitaker, C. M.; Kott, K. L.; McMahon, R. J. *J. Org. Chem.* **1995**, 60, 3499. (e) Caron, L.; Canipelle, M.; Tilloy, S.; Bricout, H.; Monflier, E. *Tetrahedron Lett.* **2001**, 42, 8837. (f) Brune, H. A.; Falck, M.; Hemmer, R.; Schmidtberg, G.; Alt, H. G. *Chem. Ber.* **1984**, 117, 2791.
- (2) (a) Letsinger, R. L.; Nazy, J. R.; Hussey, A. S. *J. Org. Chem.* **1958**, 23, 1806. (b) Ravindar, V.; Hemling, H.; Schumann, H.; Blum, J. *Synth. Commun.* **1992**, 22, 841.
- (3) (a) Knochel, P.; Singer, R. D. *Chem. Rev.* **1993**, 93, 2117. (b) Knochel, P.; Jones, P. In *Organozinc Reagents, A Practical Approach*; Harwood, L. M.; Moody, C. J., Eds.; Oxford University Press: Oxford, **1999**.
- (4) Langer, F.; Knochel, P. *Tetrahedron Lett.* **1995**, 36, 4591.
- (5) Le Gall, E.; Troupel, M.; Nédélec, J. Y. *Tetrahedron* **2003**, 59, 7497.
- (6) Fillon, H.; Gosmini, C.; Périchon, J. *J. Am. Chem. Soc.* **2003**, 125, 3867.
- (7) In a typical procedure, about 10–12 mmol of the organozinc reagent were prepared from 15 mmol of an aryl bromide according to the method described in ref. 6. To the filtered solution containing the organozinc reagent was added chlorodiphenylphosphine (10 mmol) at r.t. Stirring was continued for additional 2 h at r.t. The reaction was quenched using a 5% HCl solution and the resulting mixture was extracted with CH_2Cl_2 . After evaporation to dryness, the chromatographic purification of the crude oil over silica gel using a pentane– Et_2O mixture as an eluent afforded the analytically pure phosphine derivative.
- (8) The protection of phosphines could be verified using ^{31}P NMR (80 MHz, CDCl_3). Indeed, the δ value shifts from ca. –12 ppm (R_3P) to ca. 13 ppm (R_3PBH_3).
- (9) The original procedure: Imamoto, T.; Kusumoto, T.; Suzuki, N.; Sato, K. *J. Am. Chem. Soc.* **1985**, 107, 5301. This procedure, which reported the concomitant reduction–protection of phosphine oxides, was simplified according to the fact that no reduction was necessary. Consequently LiAlH_4 has been removed from the procedure.
- (10) In order to add exactly the required amount of the organozinc compound, a titration was realized as follows: aliquots of the solution were exposed to iodine and analyzed using GC. The amount of aryl iodide so obtained was compared to the amount of the starting aryl bromide using dodecane as internal standard.
- (11) Coupling products were characterized using ^1H NMR (200 MHz, CDCl_3), ^{13}C NMR (50 MHz, CDCl_3) when required, ^{31}P NMR (80 MHz, CDCl_3), mass spectroscopy and, when useful, ^{19}F NMR (188 MHz, CDCl_3) and FT-IR spectroscopy.

Some Data for Selected Compounds.

Diphenyl[3-(trifluoromethyl)phenyl]phosphine (**1d**): ^1H NMR: δ = 7.80–7.10 (m, 14 H) ppm. ^{31}P NMR: δ = –9.99 ppm. ^{19}F NMR: δ = –62.38 ppm. MS: m/z (rel. intensity) = 330 (100) [M], 251 (19) [M – 79], 203 (18) [M – 127], 183 (48) [M – 147], 108 (22) [M – 222].

Phenylbis[3-(trifluoromethyl)phenyl]phosphine (**2c**): ^1H NMR: δ = 7.84–7.23 (m, 13 H) ppm. ^{31}P NMR: δ = –9.97 ppm. ^{19}F NMR: δ = –62.71 ppm. MS: m/z (rel. intensity) = 399 (24) [M + 1], 398 (100) [M], 397 (21) [M – 1], 251 (40) [M – 147], 203 (44) [M – 195], 183 (22) [M – 215].

Tris(3-methylphenyl)phosphine (**3g**): ^1H NMR: δ = 7.15–6.94 (m, 12 H), 2.19 (s, 9 H) ppm. ^{31}P NMR: δ = –10.22 ppm. MS: m/z (rel. intensity) = 304 (100) [M], 303 (37) [M – 1], 211 (36) [M – 93], 197 (20) [M – 107].

(4-Methoxyphenyl)(phenyl)(o-tolyl)phosphine oxide (**4c**): ^1H NMR: δ = 7.83–6.92 (m, 13 H), 3.81 (s, 3 H), 2.43 (s, 3 H) ppm. ^{31}P NMR: δ = 29.08 ppm. MS: m/z (rel. intensity) = 322 (40) [M], 321 (100) [M – 1], 213 (21) [M – 109].